Local Income Taxes and Residential Sorting
Evidence from LEHD Origin–Destination Data

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Motivation

- Tiebout (1956): households "vote with their feet" across local jurisdictions offering different tax/service bundles.
- Strong evidence at the state level (**??**) and for sales tax cross-border shopping (**?**).
- But little causal evidence on local income tax sorting within metros — where moving costs are small and boundaries are frequent.
- Two data problems: local tax jurisdictions are sub-county (IRS/ACS too coarse), and credit systems + enforcement complicate effective tax burden.

This Paper

- Question: Do local income tax rate changes cause residential sorting across municipal boundaries?
- Setting: Pennsylvania, 2002–2019. Universal Earned Income Tax with continuous rate variation across ∼2,500 munis × ∼500 school districts.
- Data: LEHD LODES origin–destination, linking residence tract ↔ workplace tract annually.
- Outcome: `share_live_in` = (workers who live and work in city i) / (total workers in i), plus count outcomes (resident workers, total jobs).
- Identification: Act 32 of 2008 (effective Jan. 1, 2012) consolidated collection and made employer withholding mandatory — a clean enforcement break in the middle of the panel.

Institutional Setting (Pennsylvania)

- Universal EIT under Act 511 (1965); ∼2,500 munis and ∼500 school districts each set rates. Typical 0.5% muni + 0.5% SD. 291 muni increases, 162 decreases in panel.
- Each address has a six-digit **PSD code** = (muni × SD) identifying its taxing jurisdictions for withholding.
- Act 32 of 2008 (eff. Jan. 1, 2012): consolidated collection into 69 Tax Collection Districts; mandated employer withholding at the higher of home-resident vs. work-nonresident rate. Before 2012, withholding fragmented and nonresident registry incomplete.
- **Three rates per muni**: $\tau^{\text{res,muni}}$ (residents), $\tau^{\text{nr,muni}}$ (nonresidents working there), τ^{SD} (residents only – SDs do not levy on nonresidents).

How the PA EIT works (post-Act 32, ex. Philadelphia)

Worker bill = higher of *total home rate* (muni res + SD) and *work-location nonres rate*. Split between cities is piecewise:

$$\left(\tau^{\text{home}},\ \tau^{\text{work}}\right) = \begin{cases} \left(\tau_j^{\text{res}} + \tau_j^{\text{SD}} - \tau_i^{\text{nr}},\ \tau_i^{\text{nr}}\right) & \text{if } \tau_j^{\text{res}} + \tau_j^{\text{SD}} > \tau_i^{\text{nr}} \\ \left(0,\ \tau_i^{\text{nr}}\right) & \text{if } \tau_j^{\text{res}} + \tau_j^{\text{SD}} \leq \tau_i^{\text{nr}} \end{cases}$$

- Work jurisdiction always collects its full nonresident rate.
- Home muni + SD share what's left, zero if work rate is higher.
- Which rate binds determines the response margin: *home-binds* ⇒ move home; *work-binds* ⇒ change job (or absorb).

Worked example: how this maps to the model

Worker 1 lives in city B (muni 1.0% + SD 0.5% = total home 1.5%) and works in city A (muni nonres 0.8%).

- Bill $= \max(1.5\%, 0.8\%) = 1.5\%$. **Home binds**. B + its SD collect $1.5 - 0.8 = 0.7$pp; A collects 0.8pp.
- In our model: this flow $B \to A$ is treated as a **home-binds observation**. Treatment intensity for residential margin $= \tau_B^{\text{home}} = 1.5\%$.
- If B raises SD by 0.5pp (total home now 2.0%), worker 1's bill rises to 2.0% – response margin is move home.
- If A instead raises nonres to 1.7%, regime flips: bill = 1.7% at A, home gets 0. **Work binds**; worker 1 can only respond by changing job.

Implication: Q2/Q3/Q4 (resident sorting) load on $\tau_B^{\text{res,muni}} + \tau_B^{\text{SD}}$. Commuter flow models (Q4b/Q6) load on τ_A^{nr} at the work side.

Philadelphia: a separate regime

- Philadelphia is **exempt from Act 32**; its Wage Tax operates under the Sterling Act (1932), administered by the City Department of Revenue.
- Philly residents pay Philly's resident rate regardless of where they work — no max rule, no offset.
- Philly nonresidents working in Philly pay Philly's nonresident rate to Philly; no residual flows back to home muni.
- PA residents who work in Philly: Act 32 does not apply; employer withholds Philly nonresident rate.
- In our analysis: kept in main sample (it's part of PA's labor-market geography), **excluded in robustness checks** because its rate dynamics and reporting differ from Act-32 munis.

Preview of Findings

- Muni rate increases reduce `share_live_in` by -1.3 share-points per 1 pp rate hike, building over 7–8 years.
- Effect is entirely post-Act 32: muni rate effect is a precise zero pre-2012, -1.8 share-points** post-2012.
- Response is asymmetric: increases push out, decreases do not bring back.
- School district rate has *opposite sign*: SD attracts residents (Q5). The home-side decomposition is essential – bundling muni + SD into a "total" rate hides both channels.
- City-wide relocation, not boundary arbitrage.
- Fiscally productive: EIT revenue +67%, total tax +30%, population/property values unaffected.

Data Sources

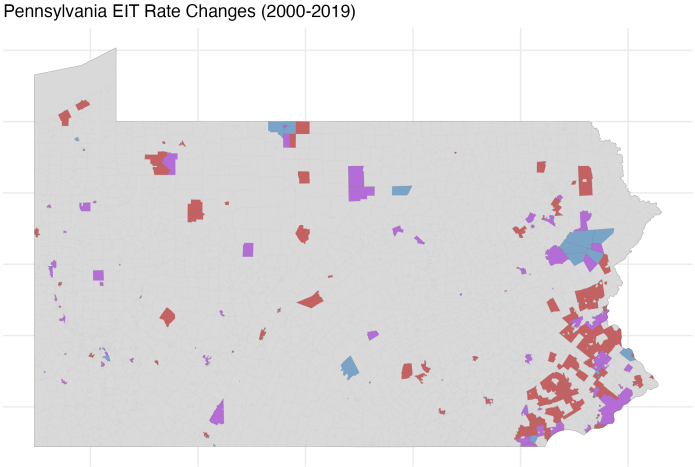
- LODES v8 (2002–2019, trimmed pre-COVID): tract-pair-year OD flows, main + aux files, segments SE01/SE03/SA01-03/JT04.
- PA rates: scraped DCED MunStats (resident + nonresident muni EIT, school district EIT) ∼2,500 munis/yr; name-based COUSUBFP crosswalk (99.6% match).
- Neighbor pairs: all PA muni pairs within 80 km centroid distance (∼1M pairs; regional commute shed).
- PA fiscal controls: DCED Annual Financial Reports (71 cols: EIT rev, police, fire, streets, sewer, parks, comm dev, debt, state aid, pop, MV).
- Crime: FBI UCR (Kaplan); Permits: Census BPS.
- April 1 alignment between rate panel and LODES reference date.

Summary Statistics (Pennsylvania, 2002–2019)

Variable	Mean	Median	SD	Max
share_live_in	0.149	0.127	0.097	0.700
Muni resident EIT (%)	0.542	0.500	0.199	4.331
Muni nonres EIT (%)	0.546	0.500	0.207	4.331
SD EIT (%)	0.523	0.500	0.221	1.000
Total jobs	3,624	1,224		
Resident workers	674	154		
Muni × year	27,973 (1,479 munis)			

Notes: Sample restricted to munis with > 50 total jobs. Pre-2012 muni nonres rates backfilled from earliest reported post-2012 value where missing (96% of munis stable).

Treatment Geography: PA Rate Changes



Red = increased; blue = decreased; purple = both; grey = unchanged.

Three Channels of Adjustment

When muni A raises its rate, workers in A can:

1. Move residence only (keep job in A). Under credit system: liability still τ_A. Zero tax savings, but reduces `share_live_in`.
2. Move residence and job to lower-rate B. New liability τ_B. Canonical Tiebout margin.
3. New hires choose to live outside A. Reduces share even without existing-resident moves.

Predictions we'll test: (1) enforcement matters $\Rightarrow$ post-Act 32 effect (Act 32 break); (2) destination rate irrelevant for commuters when work-binds; (3) complete relocators pick lower-rate destinations; (4) SD rate opposite sign (Q5); (5) boundary gradient (Q6).

Q2: Do rate increases cause sorting?

Unit i = PA municipality (cousub); t = year, 2002–2019. Outcome $\texttt{share_live_in}_{it}$ = workers who live and work in muni i at year t, divided by total workers in i. Workers responding pay home stack $\tau_i^{\text{res,muni}} + \tau_i^{\text{SD}}$; nonres rate doesn't enter. Three specs:

$$\text{(A) total stack:} \quad Y_{it} = \alpha_i + \gamma_t + \beta\, \tau_{it}^{\text{home}} + \varepsilon_{it}$$

$$\text{(B) decomposition:} \quad Y_{it} = \alpha_i + \gamma_t + \beta^{m} \tau_{it}^{\text{res,muni}} + \beta^{sd} \tau_{it}^{\text{SD}} + \varepsilon_{it}$$

$$\text{(C) muni only (back-compat):} \quad Y_{it} = \alpha_i + \gamma_t + \beta\, \tau_{it}^{\text{res,muni}} + \varepsilon_{it}$$

- Sun–Abraham + DCDH staggered estimators with three cohort definitions: muni-rate event, SD-rate event, total-stack event.
- Fiscal controls: police, fire, streets, sewer, parks, comm dev, gen admin, debt, state aid (full set).

Q2 Answer: muni rate drives the result; SD adds noise

	Muni only	Total stack	Decomp
$\tau^{\text{res,muni}}$	-0.013^{**}		-0.013^{**}
τ^{SD}			$+0.003$
τ^{home}		-0.005	

- Muni rate: -1.3 share-pp per 1 pp hike. SD rate: zero direct effect on `share_live_in`.
- Bundling them dilutes the signal – SD rate is uncorrelated with the sorting margin we measure here.
- SD does have effects on resident worker counts (Q5) with opposite sign from muni. The decomposition is essential.

Mechanism: Is it enforcement? (Act 32)

Unit i = PA municipality; t = year. Outcome $\texttt{share_live_in}_{it}$. Structural break on rate effect:

$$Y_{it} = \alpha_i + \gamma_t + \beta^{\text{pre}}\tau_{it} \cdot \mathbf{1}[t < 2012] + \beta^{\text{post}}\tau_{it} \cdot \mathbf{1}[t \geq 2012] + \varepsilon_{it}$$

- Pre-2012: fragmented collection, weak withholding; DCED's nonresident rate registry was incomplete (only ∼55% of munis reported nonres rates; jumped to ∼72% in 2012).
- Post-2012: centralized 69 TCDs, mandatory employer withholding at $\max(\tau^{\text{home}}, \tau^{\text{nr,work}})$.
- Run break for both rates separately to see which channel is enforcement-bound.

Act 32 Answer: Sorting only once the tax is binding

	Pre-2012	Post-2012
$\tau^{\text{res,muni}}$ (muni only)	-0.003	-0.018^{***}
τ^{home} (total stack)	-0.005	-0.005
$\tau^{\text{res,muni}}$ (decomp)	-0.003	-0.017^{***}
τ^{SD} (decomp)	-0.001	$+0.007$

- Muni rate: precise zero before, -1.8 share-pp after. Difference significant at 1%. Same statutory rate, very different behavioral response.
- Total stack hides the break: SD rate did not get an Act-32 enforcement shock (state-administered withholding existed already).
- Decomposition: enforcement break is on the **muni channel only**. SD plays a different role (Q5).

Q3: Are rate changes symmetric?

Unit i = PA municipality; t = year. Outcome $\texttt{share_live_in}_{it}$. Split cumulative dose from each muni's baseline (first observed) rate:

$$Y_{it} = \alpha_i + \gamma_t + \beta^+ R_{it}^+ + \beta^- R_{it}^- + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

- R_{it}^+, R_{it}^-: cumulative increases / decreases from baseline.
- If $|\beta^+| > |\beta^-| \Rightarrow$ residential choices are sticky (fixed costs of moving).
- Also estimate joint inc/dec event indicators.

Q3 Answer: Sharply asymmetric on muni rate

	R^+ (cum. inc)	R^- (cum. dec)
Muni only	-0.018^{**}	$+0.002$
Total stack	-0.006	-0.001
Decomp: muni	-0.018^{***}	$+0.001$
Decomp: SD	$+0.003$	$+0.019$

- Asymmetry on **muni rate only**: increases push out (-1.8 share-pp per pp), decreases do not bring back.
- SD rate: no asymmetry on share_live_in (SD effect lives on count outcomes – Q5).
- Policy: a city that raises then lowers its muni rate will not recover the residents it lost.

Q4: Where do displaced residents go?

Two units, two specs:

- **(a) Resident workers**: unit B = PA municipality; t = year. Outcome = log(workers living in B, any workplace). Full panel; home channel only.

$$\ln(\text{ResW}_{Bt}) = \alpha_B + \gamma_t + \beta_1 \tau_{Bt}^{\text{home}} + \beta_2 \bar{\tau}_{Bt}^{\text{nbr}} + \varepsilon$$

- **(b) Commuter flow**: unit = ordered muni-pair (A, B) with $A \neq B$, within 80 km; t = year. Outcome = log(workers commuting from home B to work A). Post-Act-32 only.

$$\ln(\text{Flow}_{ABt}) = \alpha_{AB} + \gamma_t + \beta_W \tau_{At}^{\text{nr}} + \beta_H \tau_{Bt}^{\text{home}} + \varepsilon$$

Predictions: $\beta_2 > 0$ in (a) – neighbors absorb displaced residents; $\beta_H \approx 0$ in (b) when work binds (credit system makes home rate moot for commuters).

Q4 Answer: Residents leave; neighbors absorb

	log(Resident Workers) home stack on B, full panel	Commuter flow $B \to A$ post-2012, piecewise
Own muni $\tau_B^{\text{res,muni}}$	-0.122^{**}	$+0.054^{***}$
Own SD τ_B^{SD}	$+0.147^{***}$	≈ 0
Neighbor avg	$+1.06^{*}$	—
Work nonres τ_A^{nr}	—	$+0.019$ (n.s.)

- Resident workers in B: muni rate repels (-12% per pp); SD rate attracts ($+15\%$). *Neighbors absorb* the displaced ($\beta = +1.06^{*}$).
- Neighbor = within 80 km centroid distance (regional commute shed).
- Commuter flow $B \to A$: work nonres rate near zero (small, n.s.). Home muni rate positive on flow – a *selection signature*, not causal: high-rate home cities house residents with stronger commuting margins.
- Both panels much stronger post-Act-32 (resident-workers muni effect $5\times$ larger).

Q5: Muni vs SD — decomposition across outcomes

Unit i = PA municipality; t = year. Decomposition $\beta^{m}\tau_{it}^{\text{res,muni}} + \beta^{sd}\tau_{it}^{\text{SD}}$ estimated for each of four outcomes. Q4 looked at one outcome (resident worker count) with neighbor exposure; Q5 fixes the decomposition spec and varies the outcome to identify which workers respond to which rate.

- Outcomes: `share_live_in`, log(resident workers), log(nonresident workers), log(total jobs).
- Theory: SD only taxes residents $\Rightarrow \beta^{sd} \approx 0$ on nonresident outcome but nonzero on resident outcomes. Muni res rate same direction across all but smaller for nonres workers (who pay only $\tau^{\text{nr,muni}}$, which is correlated but not identical).
- Muni–SD rate correlation $r = -0.12 \Rightarrow$ independent variation.

Q5 Answer: Opposite signs — SD attracts residents

	share_live_in	log(Resident)	log(Nonres.)	log(Total)
$\tau^{\text{res,muni}}$	-0.010^{**}	-0.117^{**}	-0.078^{*}	-0.094^{**}
τ^{SD}	$+0.002$	$+0.141^{***}$	$+0.077$	$+0.076$
Total stack	~ 0 on all – hides both channels			

- Muni rate consistently negative; SD strongly positive on residents, ~ 0 on nonresidents (theory: SDs don't tax commuters).
- Bundling them as "total EIT" nets to zero $\Rightarrow$ prior literature using a single rate would find a null.
- Decomposition is the methodological contribution.

Robustness

- Pre-trends clean across 11 observables: revenue, EIT rev, RE tax, streets, pop, debt, jobs, crime, permits. Exceptions (police, market value) addressed with controls / coverage notes.
- Fiscal controls (log police, streets, debt, state aid) leave point estimates and SEs essentially unchanged.
- Philadelphia exclusion: results robust to dropping Philly.
- Flip-floppers: 37% of PA treated munis flip-flop; excluding them strengthens results.
- Placebo: Federal workers (JT04) show no response (federal wages not subject to local withholding same way).
- ACS / IRS SOI: county-level migration null — LODES detects what coarser data cannot.

Putting it Together (1/2): Who moves?

Setup. A post-Act 32 PA city: 1,000 total workers; 150 live-and-work there (share_live_in = 15%); EIT rises 0.5% → 1.5%.

Q2/Q3. Over 7–8 years, share falls to ∼13.2% ⇒ ∼18 of 150 locally-resident workers gone. If the city later cuts the rate back, none of them return.

Q4. Those 18 leave the city entirely (not a neighborhood move); nearby lower-rate munis within 80 km absorb them.

Q5. If the overlapping school district had raised its rate instead, residents would have moved in, not out.

Q6. The loss is spread across the city, not concentrated at municipal boundaries — no "edge effect."

Putting it Together (2/2): Was it worth it?

Same city: 1,000 workers, rate 0.5% → 1.5%, ∼18 residents lost.

Fiscal. EIT revenue roughly doubles (+67 log points); total tax +30%. Property values and population: unchanged.

Q7. Most new dollars flow to community development (planning, econ dev, housing) — not to police patrols residents see.

nforcement. None of this happens pre-Act 32: same statutory rate, precise zero behavioral response. Collection capacity is the binding constraint.

Bottom line. The city trades ∼18 residents for a near doubling of EIT revenue. Rate increases work — when enforced.

Summary

1. Local income tax rate increases cause residential sorting: `share_live_in` falls by 0.5–0.8 share-points per 1 pp rate hike, building over 7–8 years.
2. Effect is enforcement-contingent: zero pre-Act 32, large post-2012.
3. Sorting is asymmetric (sticky residences) and city-wide (not boundary arbitrage).
4. Muni and SD rates have opposite signs — SD acts as an amenity.
5. Rate increases are fiscally productive: base erosion too small to offset revenue gains.

Next Steps

- Link Act 1 SD rate changes to actual property tax millage movements (PDE data 2015+ restrictive).
- Extend Q7 with richer, pre-treatment expenditure composition (more power on visibility channel).
- Apply the LODES-based design to other local instruments (property tax millage, local sales taxes, fees).
- Welfare: combine behavioral elasticity with revenue elasticity to bound efficiency cost of local income tax.

Thank You

Questions?

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github.com/MatthewHockert/local-income-tax-migration